Ahara – Food Redistribution & Waste Reduction Platform

System Design Document

1. Introduction

1.1 Project Overview

Ahara is a technology-driven food redistribution platform designed to reduce surplus food waste by securely connecting sellers, buyers, and volunteers. The platform ensures food safety, transparency, and traceability while optimizing pickup coordination and environmental impact tracking.

1.2 Problem Statement

Food waste contributes to environmental degradation and resource inefficiency. Surplus food often goes unused due to lack of structured redistribution systems.

Key Challenges:

- Lack of trust in redistribution systems

- Poor logistics coordination

- Limited traceability and compliance tracking

- No measurable impact transparency

1.3 System Objective

The Ahara platform aims to:

- Enable safe surplus food listing

- Enforce role-based access control

- Ensure hygiene and safety compliance

- Optimize pickup logistics

- Provide environmental and social impact visibility

- Maintain audit-ready compliance records

2. System Architecture

Figure 1: System Architecture Diagram

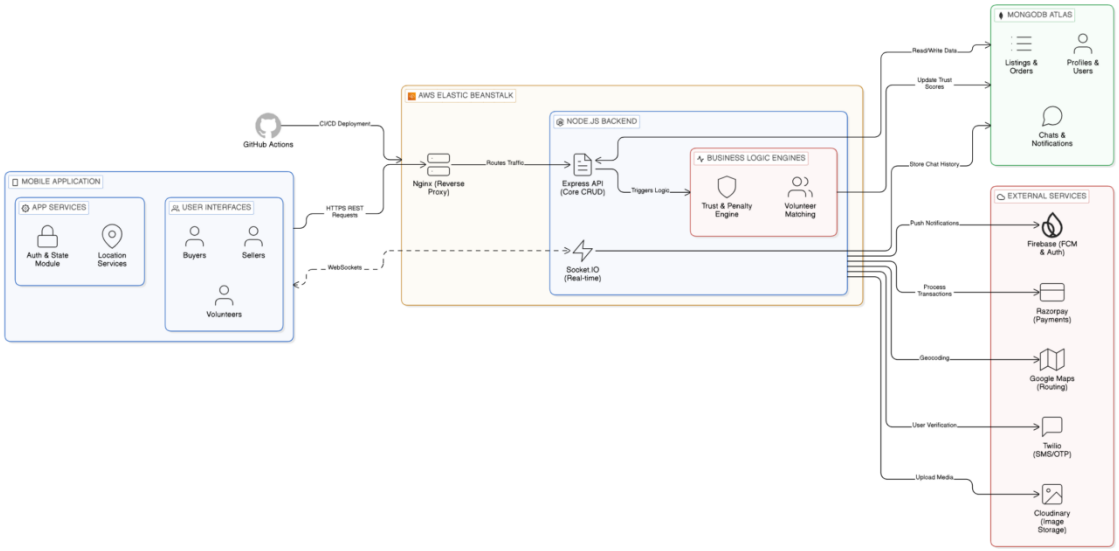


Figure 1

Presentation Layer: Flutter Mobile App, Web Interface

Application Layer: Node.js + Express backend, RBAC middleware

Data Layer: MongoDB, Firebase Authentication

External Services: Google Sign-In, Maps API, Notifications

3. Use Case Diagram

Figure 2: Use Case Diagram

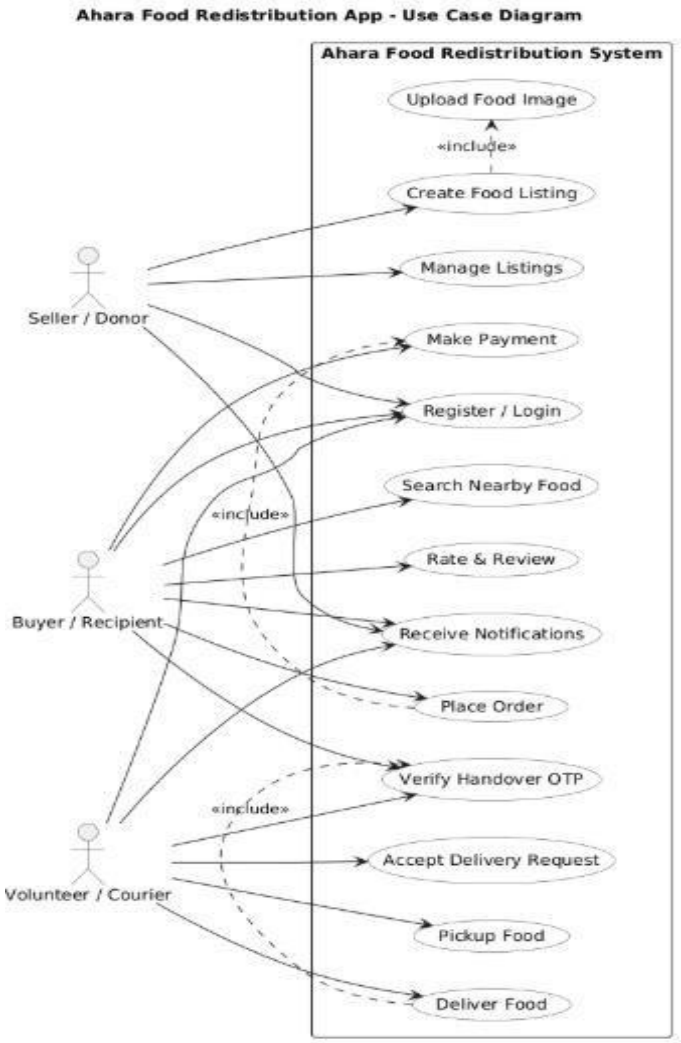


Figure 2

Actors:

Buyer – Browse listings, request pickup, view impact

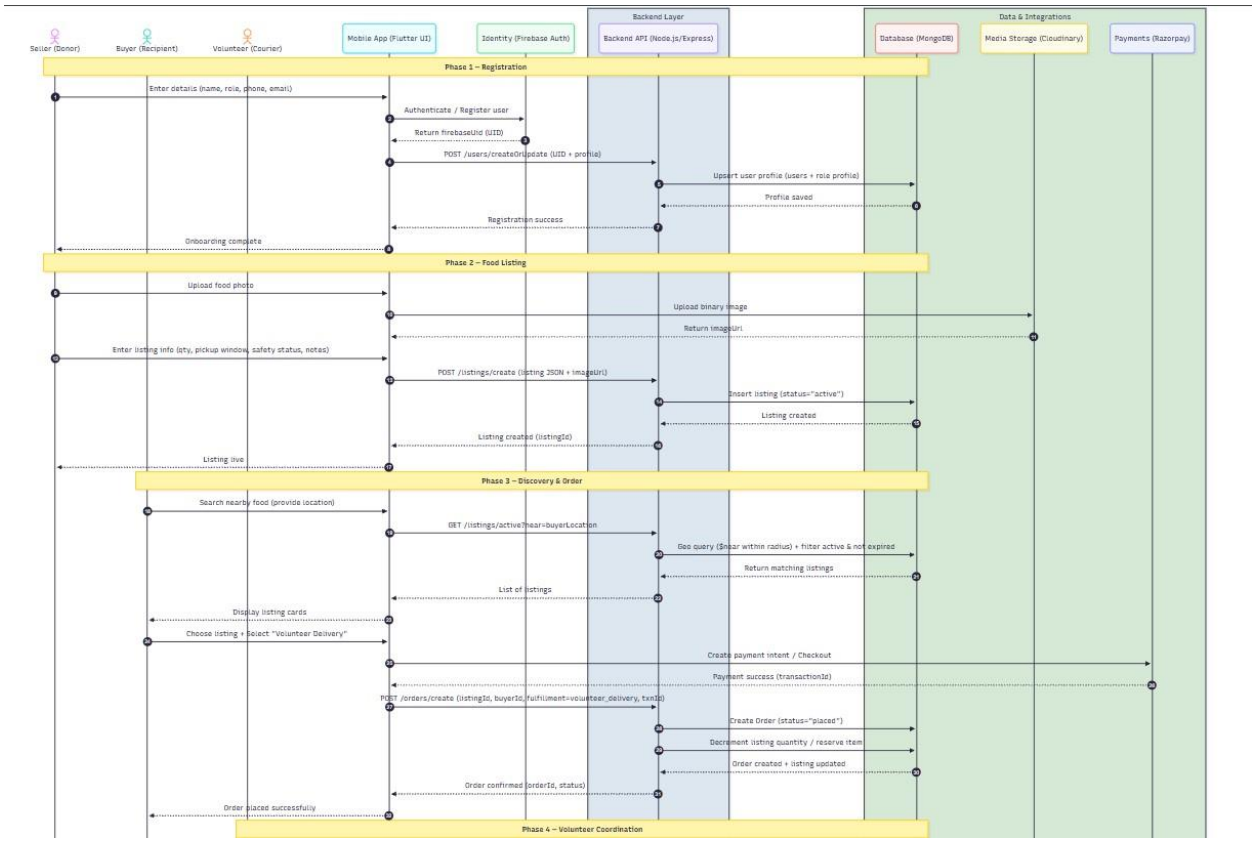
Seller – Create listings, submit hygiene declaration

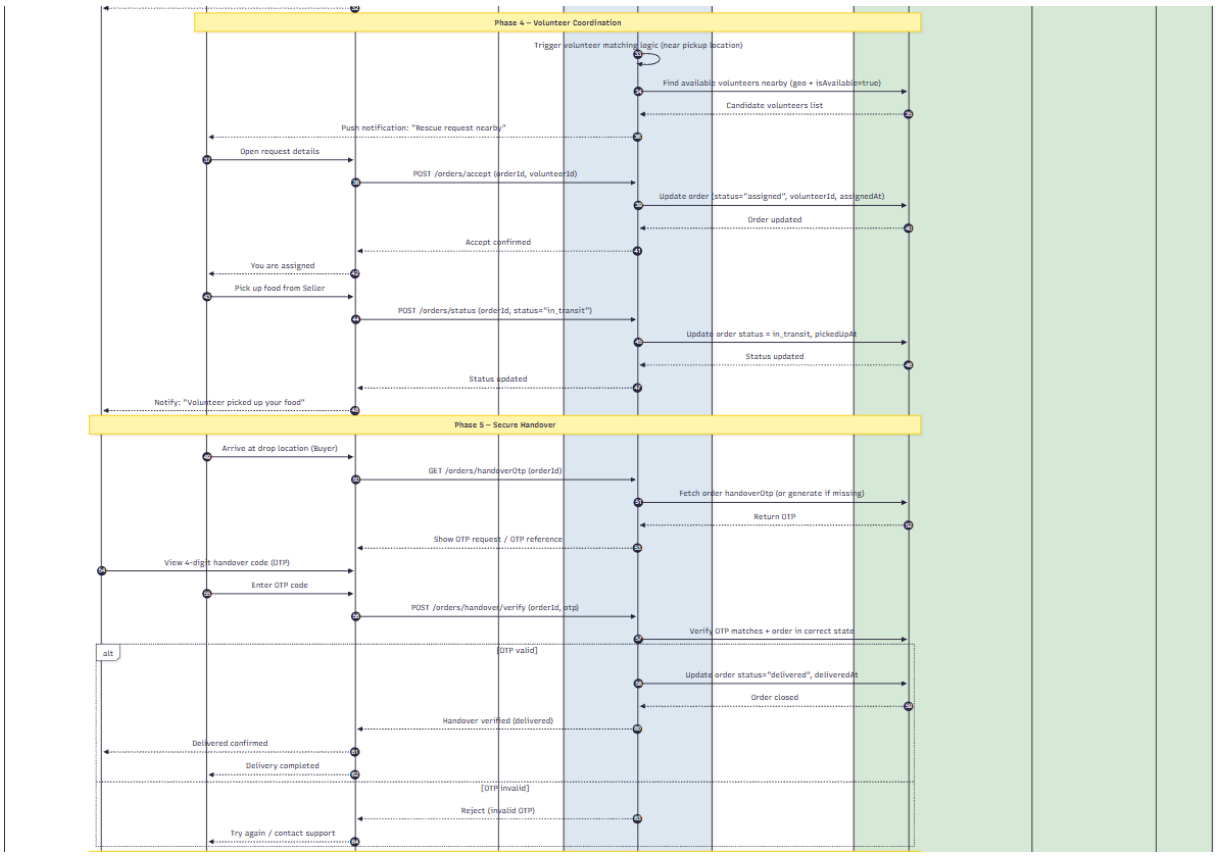
Volunteer – Accept pickups, update status

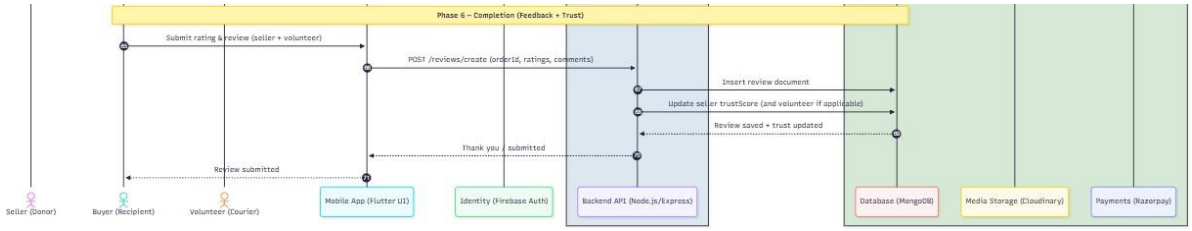
Admin – Manage users, compliance, reports

4. Sequence Diagram

Figure 3: Sequence Diagram




Phase 4 – Volunteer Coordination
Trigger volunteer matching logic (near pickup location)
Find available volunteers nearby (geo + isAvailable=true)
Candidate volunteers list
Push notification: "Rescue request nearby"
Open request details
POST /orders/accept (orderId, volunteerId)
Update order status="assigned", volunteerId, assignedAt)
Order updated
Accept confirmed
You are assigned
Pick up food from Seller
POST /orders/status (orderId, status="in_transit")
Update order status = in_transit, pickedUpAt
Status updated
Status updated
Notify: "Volunteer picked up your food"
Phase 5 – Secure Handover
Arrive at drop location (Buyer)
GET /orders/handoverOtp (orderId)
Fetch order handoverOtp (or generate if missing)
Return OTP
Show OTP request / OTP reference
View 4-digit handover code (OTP)
Enter OTP code
POST /orders/handover/verify (orderId, otp)
Verify OTP matches + order in correct state
alt
[OTP valid]
Update order status="delivered", deliveredAt
Order closed
Handover verified (delivered)
Delivered confirmed
Delivery completed
[OTP invalid]
Reject (invalid OTP)
Try again / contact support


Phase 6 – Completion (Feedback + Trust)
Submit rating & review (seller + volunteer)
POST /reviews/create (orderId, ratings, comments)
Insert review document
Update seller trustScore (and volunteer if applicable)
Review saved + trust updated
Thank you / submitted
Review submitted
Seller (Donor)
Buyer (Recipient)
Volunteer (Courier)
Mobile App (Flutter UI)
Identity (Firebase Auth)
Backend API (Node.js/Express)
Database (MongoDB)
Media Storage (Cloudinary)
Payments (Razorpay)

Flow: User submits registration → Firebase Auth → Backend JWT verification → MongoDB storage → Confirmation response.

Flow: Seller submits listing → Backend validation → Safety checks → MongoDB storage → Matching trigger.

5. Schema Diagram

Figure 4:

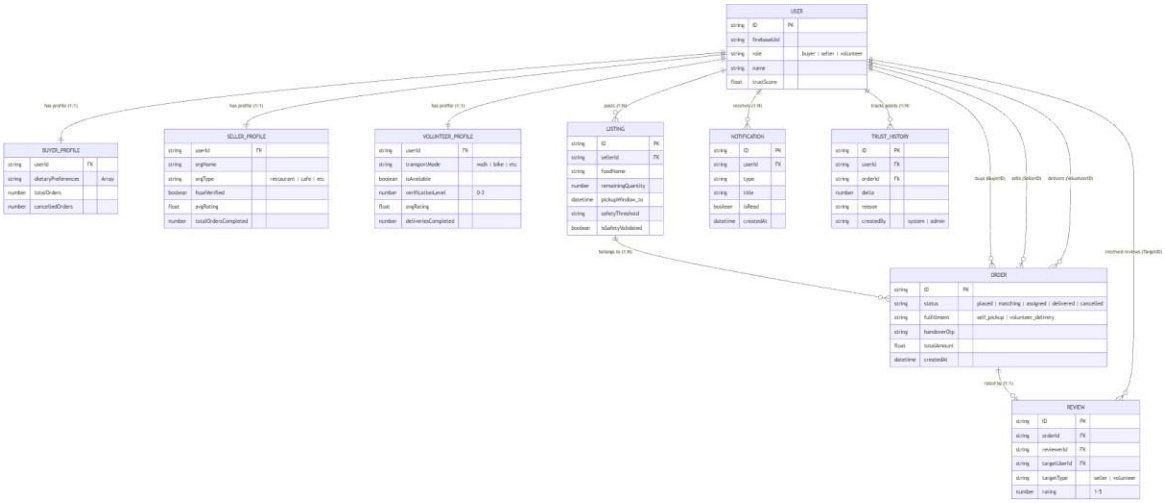


Figure 4

6. Class Diagram

Figure 5: Class Diagram

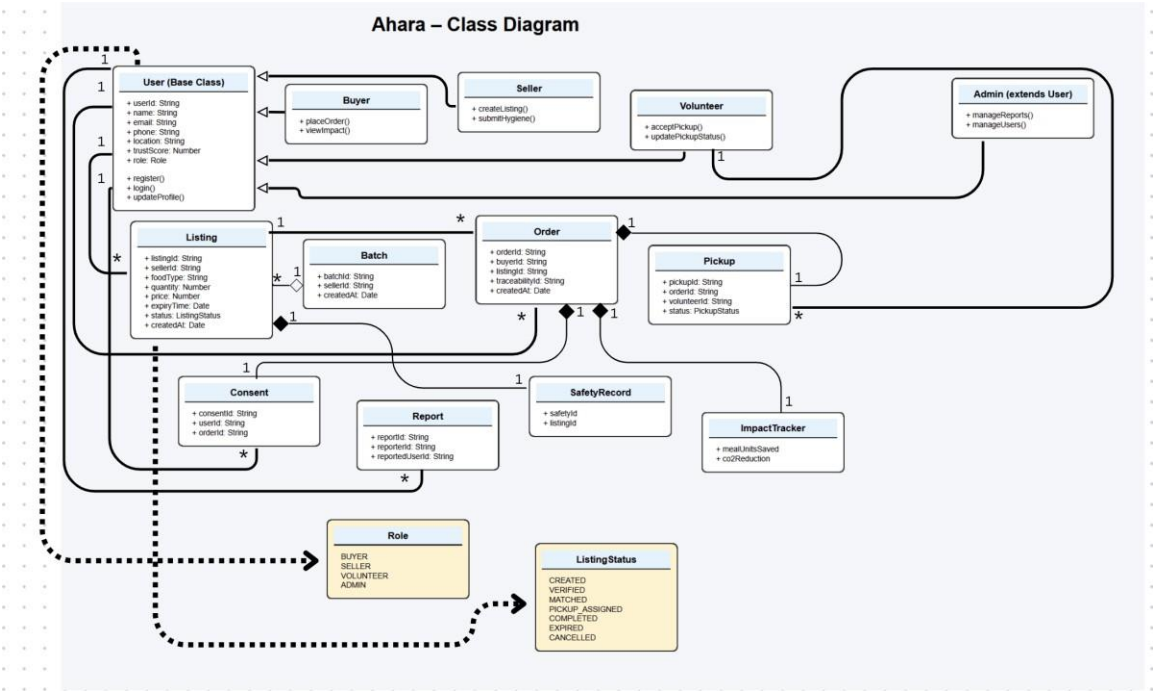


Figure 5

7. Conclusion & Future Enhancements

The Ahara platform integrates identity management, safety enforcement, logistics optimization, and impact tracking into a unified architecture.